



Chapter 12 – DoS (Denial of Service) & DDoS (Distributed Denial of Service) Attacks



Introduction to DoS Attack



What Does "Denial of Service" Mean?

- **Denial** = Rejection ❌
- **Service** = Service provided by a server 🌐

A **Denial of Service (DoS)** attack happens when an attacker sends so much unwanted traffic to a server that legitimate users cannot access its services.



Real-Life Example

Imagine:



A restaurant has:

- 20 tables
- Capacity for 100 customers

Now one person sends **1000 fake customers** to occupy all tables.

Result:

- ❌ Real customers cannot sit
- ❌ Restaurant becomes unavailable

The restaurant is still running, but genuine customers cannot use it.

This is exactly how a DoS attack works.

Other Example Explained

Suppose:

 Server Capacity = 1 GB incoming traffic

 Attacker sends = 10 GB unwanted traffic

What happens?

1. Server starts processing incoming requests
2. Server can only handle 1 GB
3. Remaining traffic fills resources
4. CPU becomes busy
5. RAM usage increases
6. Bandwidth becomes saturated

Result:

- ✗ New users cannot access the service
- ✗ Website becomes slow
- ✗ Website may completely crash

This is called a **Denial of Service (DoS) Attack**.

What is a DoS Attack?

Definition

A **DoS Attack (Denial of Service)** is a cyberattack where a single machine floods a target system with excessive traffic to make it unavailable for legitimate users.

Purpose of DoS Attack

Attackers may perform DoS attacks to:

- Crash websites 
- Make services unavailable 

- Cause financial loss 💰
 - Disrupt business operations 🏢
 - Damage reputation 📉
-

📌 Example

One attacker sends:

- Millions of HTTP Requests
- Thousands of Ping Requests
- Large Amounts of TCP Packets

toward a web server.

The server becomes overloaded and stops responding.

⚡ What is a DDoS Attack?

🧠 **DDoS = Distributed Denial of Service**

A **DDoS attack** is similar to DoS, but instead of one computer attacking, thousands or millions of infected computers attack simultaneously.

🤖 What is a Botnet?

A **Botnet** is a network of infected devices controlled by an attacker.

These devices are called:

- Bots 🤖
- Zombies 🧟

Examples:

- Computers
- Laptops
- Mobile phones

- Smart TVs
- CCTV Cameras
- IoT Devices



Difference Between DoS and DDoS

Feature	DoS	DDoS
Number of Attackers	One	Thousands
Traffic Volume	Lower	Very High
Detection	Easier	Difficult
Blocking	Easy	Hard
Impact	Medium	Very Severe



How DDoS Works

Step 1: Malware Infection

Attacker infects many devices using:

- Trojans
- Worms
- Bots

Step 2: Command & Control Server

All infected devices connect to a:



Command & Control (C&C) Server

This server controls all bots.

Step 3: Attack Command

Attacker sends one command:

"Attack this website"

Step 4: Traffic Flood

Thousands of bots start sending:

- HTTP Requests
- TCP SYN Packets
- UDP Packets
- ICMP Ping Packets

towards the target.

Step 5: Service Down

Server resources become exhausted:

- CPU = 100%
- RAM Full
- Network Bandwidth Saturated

Result:

🚫 Website becomes unavailable

📦 Types of Packets Used in DDoS

◆ ICMP Packets

Used in ping requests.

Example:

ping google.com

◆ TCP SYN Packets

Used during TCP Handshake.

Can be abused in:

SYN Flood Attacks

◆ UDP Packets

Connectionless packets.

Often used for amplification attacks.

◆ HTTP Requests

Attackers repeatedly request:

/index.html

/login.php

making web servers overloaded.



Tool: hping3

What is hping3?

hping3 is a packet crafting and packet analysis tool.

It can generate:

- TCP Packets
 - UDP Packets
 - ICMP Packets
 - Custom Packets
-

Why Use hping3?

Ethical Hackers use it to:

- ✓ Test firewalls
- ✓ Analyze networks
- ✓ Simulate DoS attacks in labs
- ✓ Perform packet crafting

Cyber Attack News Website

Bleeping Computer

You can read:

- Latest cyberattacks
- Ransomware incidents
- Data breaches
- Malware campaigns

to stay updated with cybersecurity trends.

Important hping3 Commands

1 Send SYN Packets

hping3 <target_ip>

Default behavior sends TCP SYN packets.

2 Fast Mode

hping3 --fast <target_ip>

Sends packets faster than normal.

3 ICMP Ping Packets

hping3 -1 <target_ip>

-1 = ICMP Mode

Used to send ping packets.

4 UDP Packets

hping3 -2 <target_ip>

-2 = UDP Mode

Sends UDP packets.

5 SYN Packet to Specific Port

hping3 -S <target_ip> -p 80

Explanation

- -S = SYN Flag
 - -p 80 = Target Port
-

6 Faster SYN Flood

hping3 -S --faster <target_ip> -p 80

Generates heavy traffic.

7 Port Scanning

hping3 --scan 1-100 <target_ip>

Scans ports 1 to 100.

8 ACK Packets

```
hping3 -A <target_ip> -p 80
```

Sends ACK packets.

9 FIN Packets

```
hping3 -F <target_ip> -p 80
```

Sends FIN packets.

Used to finish TCP connections.

10 Source IP Spoofing

```
hping3 -a 2.2.2.2 <target_ip>
```

What happens?

Target sees traffic as if it came from:

2.2.2.2

instead of the actual sender.

1 1 Random Source IP

```
hping3 --rand-source <target_ip> -p 80
```

Uses random spoofed IP addresses.

1 2 Send Limited Packets

```
hping3 -c 1000 <target_ip>
```

-c = Count

Sends only 1000 packets.

1 3 Packet Interval

```
hping3 -i u1000 <target_ip>
```

Meaning

- u = Microseconds

Controls packet timing.

1 4 Flood Mode

```
hping3 --flood -S <target_ip> -p 80
```

Sends packets continuously at maximum speed.

⚠ Use only in authorized lab environments.

Understanding hping3 Parameters

Parameter	Meaning
-1	ICMP Mode
-2	UDP Mode
-S	SYN Flag
-A	ACK Flag
-F	FIN Flag
-c	Packet Count
-p	Port
-i	Interval
--fast	Fast Mode
--faster	Faster Mode
--flood	Flood Mode

Parameter	Meaning
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-a	Spoof IP
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--rand-source	Random IP
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Observing Packets with Wireshark

You can open:

Wireshark

on the target machine.

Wireshark allows you to see:

- ICMP Packets
- TCP SYN Packets
- UDP Packets
- Spoofed Traffic

in real-time.



Blocking ICMP Packets

Since ICMP is often used in ping floods, administrators can block it.

Linux Example:

```
iptables -A INPUT -p icmp --icmp-type echo-request -j DROP
```

What it does?

Drops incoming ping requests.

DDoS Prevention & Mitigation

1 Strong Firewall Rules

Firewalls can:

- ✓ Filter malicious traffic
 - ✓ Block spoofed IPs
 - ✓ Detect abnormal requests
-

2 Load Balancing

Traffic is distributed among multiple servers.

Example:

Instead of:

1 Server

Use:

Server 1

Server 2

Server 3

Server 4

This prevents overload.

3 Cloud-Based Protection

Popular services:

- [Cloudflare](#)
- [AWS Shield](#)
- [Akamai](#)

These absorb and filter attack traffic.

Rate Limiting

Restricts:

How many requests one IP can send per second

Prevents abuse.

5 IDS/IPS Systems

IDS

Intrusion Detection System

IPS

Intrusion Prevention System

Can detect:

- Traffic spikes
- SYN floods
- Bot behavior

and automatically block them.

6 Network Redundancy

Use servers in multiple regions.

Example:

 India

 USA

 Europe

If one region is attacked, others continue working.

7 DDoS Mitigation Solutions

Popular tools:

- Cloudflare DDoS Protection
- AWS Shield Advanced
- Akamai Kona
- Arbor Networks APS
- Imperva Incapsula



Lecture Summary

Topic	Description
DoS	Single machine floods target
DDoS	Multiple systems attack target
Botnet	Collection of infected devices
hping3	Packet crafting and testing tool
ICMP Flood	Ping-based attack
SYN Flood	TCP handshake abuse
UDP Flood	Massive UDP traffic
Spoofing	Hiding real source IP
Wireshark	Packet monitoring
Firewall	First line of defense
Cloudflare	DDoS protection service
IDS/IPS	Attack detection and prevention

Protection Against All Attacks Covered So Far

Password Attacks

Protection

- ✓ Strong Passwords (12–16+ characters)
 - ✓ Password Manager
 - ✓ Multi-Factor Authentication (MFA)
 - ✓ Account Lockout Policy
 - ✓ Disable Default Passwords
-

Wi-Fi Attacks

Protection

- ✓ WPA3 Security
 - ✓ Disable WPS
 - ✓ Strong Wi-Fi Password
 - ✓ Router Firmware Updates
 - ✓ Guest Network Separation
 - ✓ VPN on Public Wi-Fi
-

Nmap Scanning & Reconnaissance

Protection

- ✓ Close Unused Ports
 - ✓ Firewall Rules
 - ✓ IDS/IPS Monitoring
 - ✓ Disable Unnecessary Services
-

Metasploit Exploitation

Protection

- ✓ Regular Security Patching
 - ✓ Vulnerability Scanning
 - ✓ Endpoint Protection
 - ✓ Least Privilege Principle
 - ✓ Application Whitelisting
-

Virus Attacks

Protection

- ✓ Antivirus Software
 - ✓ Don't Download Unknown Files
 - ✓ Scan USB Devices
 - ✓ Keep OS Updated
-

Worm Attacks

Protection

- ✓ Patch Systems
 - ✓ Close Unused Ports
 - ✓ Network Segmentation
 - ✓ Updated Antivirus
-

Ransomware

Protection

- ✓ Offline Backups
 - ✓ Email Security
 - ✓ Patch Management
 - ✓ Ransomware Protection Tools
-

Bot Attacks

Protection

- ✓ EDR/XDR Solutions
 - ✓ Monitor Outbound Connections
 - ✓ Patch IoT Devices
 - ✓ Strong Authentication
-

Trojan Attacks

Protection

- ✓ Download Software from Official Sources
 - ✓ Verify Digital Signatures
 - ✓ Antivirus Protection
-

Spyware

Protection

- ✓ Anti-Spyware Tools
- ✓ Permission Management
- ✓ Avoid Suspicious Apps



Phishing Attacks

Protection

- ✓ Verify URLs
 - ✓ Verify Email Sender
 - ✓ Security Awareness Training
 - ✓ MFA
-



Spoofing Attacks

Protection

- ✓ SPF, DKIM, DMARC
 - ✓ Caller Verification
 - ✓ Email Verification
-



MITM (Man-in-the-Middle)

Protection

- ✓ HTTPS
 - ✓ VPN
 - ✓ Certificate Validation
 - ✓ Avoid Public Wi-Fi
-



SQL Injection

Protection

- ✓ Prepared Statements
- ✓ Parameterized Queries

- ✓ Input Validation
 - ✓ WAF (Web Application Firewall)
-

XSS (Cross-Site Scripting)

Protection

- ✓ Input Sanitization
 - ✓ Output Encoding
 - ✓ Content Security Policy (CSP)
-

Social Engineering

Protection

- ✓ Security Awareness Training
 - ✓ Verify Requests
 - ✓ Never Share OTPs
 - ✓ Verify Identity Before Acting
-

DDoS Attacks

Protection

- ✓ Cloudflare
 - ✓ Load Balancers
 - ✓ IDS/IPS
 - ✓ Rate Limiting
 - ✓ Traffic Filtering
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